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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
Euthanasia Agents: Mice were euthanized by an i.p. injection of Zoletil (50 mgkg) and Rompun (xylazine , 10 mgkg) or by inhalation of isoflurane (3-5) , followed by cervical dislocation (32) .
IACUC: Ethics statement All animal experiments were performed in accordance with the Guide for the Care and Use of Laboratory Animals published by National Institutes of Health (publication no. 85-23 , revised 1996) and were carried out according to the protocol approved by the Animal Research Committee of Chang Gung Memorial Hospital
Field Sample Permit: Human sample collection was conducted in accordance with the Declaration of Helsinki and was approved by the Chang Gung Memorial Hospital Research Ethics Committee (IRB No. 96-1748B).
IRB: Human sample collection was conducted in accordance with the Declaration of Helsinki and was approved by the Chang Gung Memorial Hospital Research Ethics Committee (IRB No. 96-1748B).
Consent: Written informed consent was obtained from all patients.
Inclusion and Exclusion Criteria
The inclusion criteria were as follows: Adults and children with AML with MLL-t .
Attrition
Among the 114 cases with MLL-t AML (56 men and 58 women , age range 0-84 years , median 27 years) , it was determined that eight cases with MLLAF10 or MLLAF9 had sufficient remaining specimens available for HOXA11 expression level analysis .
Sex as a biological variable
Among the 114 cases with MLL-t AML (56 men and 58 women , age range 0-84 years , median 27 years) , it was determined that eight cases with MLLAF10 or MLLAF9 had sufficient remaining specimens available for HOXA11 expression level analysis .
Subject Demographics
Age: Among the 114 cases with MLL-t AML (56 men and 58 women , age range 0-84 years , median 27 years) , it was determined that eight cases with MLLAF10 or MLLAF9 had sufficient remaining specimens available for HOXA11 expression level analysis .
Weight: In vivo leukemogenesis The in vivo leukemogenic potential of the leukemia cell lines was determined by BM transplantation assay using male B6 mice (n=45 , age , 6-8 weeks weight , 20-24 g) .

Randomization	
not detected.	
Blinding	
not detected.	
Power Analysis	
not detected.	
Replication	
not required.	
Cell Line Authentication	
not detected.	
Data Information	
Availability: Microarray data are available at the NCBI Gene Expression Omnibus (GEO) website (https:www.ncbi.nlm.nih.govgeo accession nos .	
Identifiers: These different cell types were either generated in the current study (AKw1G) or in previous studies (AK2G, AK3G, APw1 and APm1) (21,22,26) (Fig. 1A).	21,22,26
Identifiers: GSE82156 and GSE134586) or can be downloaded from Chang Gung University website (21)	GSE82156
	GSE134586

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
The membrane was blocked in 5 bovine serum albumin at 4degC (Sigma-Aldrich Merck KGaA) for 1 h , and then incubated with primary antibodies at 4degC overnight against mouse Hoxa11 (1:5,000 cat. no.	Hoxa11		
NBP1-80228 Novus Biologicals , Ltd.) , b-actin or Gapdh (1:10,000 cat. nos . sc-47778 or sc-32233 Santa Cruz Biotechnology , Inc .) , followed by incubation for 2 h at room temperature with horseradish peroxidase-conjugated anti-mouse or anti-rabbit IgG secondary antibodies (1:5,000 cat. nos . C04001 or C04003 Croyez Bioscience Co . , Ltd.) .	Gapdh		
	anti-mouse		
	anti-rabbit IgG		
	C04001		
For immunophenotypic analysis , cells were stained at 4degC for 15 min with phycoerythrin-macrophage-1 antigen (Mac-1) , phycoerythrin-CD115 and allophycocyanine-Ki-67 antibodies (cat. nos . RM2804-3 , 12-1152-82 and 17-5698-82 eBioscience Thermo Fisher) followed by flow cytometric analysis using FACSCanto II Cell Analyzer and FACSDiva software version 5.0 (BD Biosciences) .	antigen (Mac-1)		
	phycoerythrin-CD115		
	allophycocyanine-Ki-67		
Experimental Models: Cell Lines			
Plasmid DNAs of pMSCV-Hoxa11 and pMSCVpuro were transfected into EcoPack2-293 cells (Invitrogen Thermo Fisher Scientific , Inc .) to package retroviruses .	EcoPack2-293		
Viral titer was determined by infection of NIH3T3 (American Type Culture Collection ATCC(r) CRL-1658 (tm)) , a murine fibroblast cell line , with serial diluted supernatant to generate puromycin-resistant colonies .	NIH3T3		Suggestion: RRID:CVCL_0594(link)
Experimental Models: Organisms/Strains			
Materials and methods Cell culture The mouse MLLAF10(OM-LZ) leukemia cell line (12G) and MLLAF10(OM-LZ) cells harboring wild-type or activating KRAS (KRASG12C) and wild-type or activating PTPN11	C57BL6J		Suggestion: RRID:IMSR_JAX:000664(link)

(PTPN11G503A) were generated by the retroviral transduction of genes to 5-fluorouracil (5-FU)-enriched C57BL6J (B6) mouse bone marrow (BM) cells.			
Recombinant DNA			
Cells stably transfected with blank lentiviral vector pLKO_025 were used as negative controls .	pLKO_025		
Plasmid DNAs of pMSCV-Hoxa11 and pMSCVpuro were transfected into EcoPack2-293 cells (Invitrogen Thermo Fisher Scientific , Inc .) to package retroviruses .	pMSCV-Hoxa11		
	pMSCVpuro		
To amplify 300-bp DNA fragments of the pMSCVpuro vector (from 12G-V) or the region spanning the pMSCVpuro-Hoxa11 junction (from 12G-H11) from genomic DNA , PCR was performed using the following primers: MSCV 5 primer 5-CCCTTGAACCTCCTCGTTCAGC C-3 in combination with MSCV 3 primer 5-GAGACGTGCTACTTCCATTTGT C-3 or Hoxa11 primer 5-GAGTAGCAGTGGGCCAGATTGC -3 .	pMSCVpuro-Hoxa11		
Software and Algorithms			
Microarray data are available at the NCBI Gene Expression Omnibus (GEO) website (https:www.ncbi.nlm.nih.govgeo accession nos .	Gene Expression Omnibus		Suggestion: Gene Expression Omnibus (GEO) (RRID:SCR_005012)(link)
A heat map was obtained using Cluster version 3.0 (http:bonsai.hgc . jpmdehoonsoftwarecluster) and Java TreeView version 1.1.6r4 (http:jtreeview . sourceforge.net) .	Cluster		Suggestion: Cluster (RRID:SCR_013505)(link)
	TreeView		Suggestion: TreeView (RRID:SCR_013503)(link)
DEGs with >1.5-fold-change in paired MLLAF10 leukemia cells harboring wild-type and oncogenic KRAS or PTPN11 were then functionally annotated with GO enrichment analysis using online Database for Annotation , Visualization and Integrated Discovery (DAVID) version 6.8 annotation tools (https:david.ncifcrf . gov) .	DAVID		Suggestion: DAVID (RRID:SCR_001881)(link)
Validation of HOXA11 expression in patients with AML To validate the differential expression and drug responsiveness of HOXA11 in	https:www.oncomine.org		

MLL-t AML , a meta-analysis was performed using the Oncomine(tm) database (https:www.oncomine.org) ,			
All lentiviruses were obtained from the National RNAi Core Facility at the Institute of Molecular BiologyGenomic Research Center , Academia Sinica (Taiwan)	Molecular BiologyGenomic		
For immunophenotypic analysis , cells were stained at 4degC for 15 min with phycoerythrin-macrophage-1 antigen (Mac-1) , phycoerythrin-CD115 and allophycocyanine-Ki-67 antibodies (cat. nos . RM2804-3 , 12-1152-82 and 17-5698-82 eBioscience Thermo Fisher) followed by flow cytometric analysis using FACSCanto II Cell Analyzer and FACSDiva software version 5.0 (BD Biosciences) .	FACSDiva		Suggestion: BD FACSDiva Software, RRID:SCR_001456(link)
Statistical analysis Statistical analyses were performed using SPSS software version 20.0 (SPSS, Inc).	SPSS		Suggestion: SPSS, RRID:SCR_002865(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Statistical significance was evaluated using Fishers exact test and corrected by Bonferroni correction for multiple testing .	Fishers exact test
	Bonferroni
The statistical significance of differences in gene expression levels of the two groups was compared using a Mann-Whitney test.	Mann-Whitney test
Survival analysis was conducted according to the Kaplan-Meier method, and differences in survival were assessed using the log-rank test.	log-rank test
Drug sensitivity was compared using an unpaired two-sample Students t-test.	t-test

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